

HNS Multi-Axis Verification PoC v1.0

Evaluating AI Output Across Generative, Structural, and Grounded Axes

Author: Satoru Hara / Natural Structure Works (NSW)

Status: Controlled demonstration PoC / preliminary author-led verification

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1. Purpose

Demonstrate that an AI output can be evaluated separately across Axis 1 generation, Axis 2 structural/causal validity, and Axis 3 grounded/executive compatibility.

2. Core Claim

The multi-axis view exposes cases where fluent generation is not equivalent to structural or grounded validity.

3. Method

This PoC uses a controlled post-hoc demonstration method. Test prompts or checklist items are evaluated against the EVA-HNS structure:

- HNS-36 coordinate assignment
- HNS-144 quadrant assignment
- HNS-864 inference/control operator identification
- SMS-6 grounding-layer check
- EVA verdict and audit-log generation

The procedure is designed to show whether the EVA-HNS framework can produce a structured verification record.

4. Result Summary

The demonstration produced an EVA audit record for every test item.

5. Interpretation

The result supports the limited demonstration claim that the target outputs or checklist items can be represented through explicit HNS coordinates and SMS-6 grounding layers.

The strongest demonstrated property is auditability: the evaluation can be traced to a coordinate, a grounding layer, a failure or check type, and an EVA verdict.

6. Limitations

This PoC does not prove the full mathematical orthogonality of the axes or validate the information-geometric hypothesis.

Additional limitations:

- This is not an independent third-party validation.
- This is not a large-scale benchmark.
- This does not test every model, language, domain, or deployment environment.
- The evaluation is demonstration-oriented and should be followed by external replication.

7. Included Files

8. Suggested Citation

Hara, S. (2026). HNS Multi-Axis Verification PoC v1.0: Evaluating AI Output Across Generative, Structural, and Grounded Axes. Natural Structure Works.

9. Status Notice

This PoC is part of the author's own research and publication program. It is not an ISO, IEC, SC42, IEEE, NIST, CEN/CENELEC, or governmental standard.

Appendix A - Detailed Result Table

test_id	test_type	hns36_lay er	hns36_cat egory	hns144_q uadrant	hns864_o perator	sms6_laye r	eva_verdi ct	grounding _score
M01	Axis 1 Pass / Axis 2 Fail	L6	C5	Objective/ Contingent	O2 vs O1 mismatch	SMS-4	flag	0.42
M02	Axis 1 Pass / Axis 3 Fail	L5	C3	Objective/ Contingent	O5 Comparato r	SMS-2/ SMS-3	flag	0.42
M03	Axis 2 Pass / Axis 3 Review	L4	C5	Objective/ Contingent	O2 Interventio n	SMS-4	review	0.63
M04	Axis 1-2-3 Pass	L4	C2	Objective/ Contingent	O1 Conditionin g	SMS-2	pass	0.92
M05	Axis 2-3 Fail	L6	C4	Subjective/ Necessary	O5 Comparato r	SMS-5/ SMS-6	flag	0.42

Appendix B - Test Set

test_id	test_type	prompt_or_item
M01	Axis 1 Pass / Axis 2 Fail	Fluent output with unsupported causal leap.
M02	Axis 1 Pass / Axis 3 Fail	Fluent output drifting from operational context.
M03	Axis 2 Pass / Axis 3 Review	Structurally clear but weakly grounded output.
M04	Axis 1-2-3 Pass	Fluent, structurally coherent, and grounded output.
M05	Axis 2-3 Fail	Category ambiguity plus governance grounding failure.

Appendix C - Audit Log Reference

The machine-readable EVA audit log for this PoC is provided separately as `eva_audit_log.json` in the same folder. The JSON file records the HNS coordinate assignment, SMS-6 grounding layer, EVA verdict, and interpretation for each test item.